KHIZAR ANJUM

Machine Learning Engineer

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PROFESSIONAL EXPERIENCE

- **Graduate Research Assistant, CPS Lab — (2019 – Present) | Piscataway, NJ, US**
  Specialized in developing efficient deep learning architectures for various applications, with focus on resource-constrained devices. Published in prestigious journals including IEEE JBHI, IEEE JSAC and conferences like IEEE PerCom, MASS, and Ucomms, resulting in multiple patents and prestigious government grant funding.
- **Research Assistant, Communications Lab — (2017 – 2019) | Lahore, Pakistan**
  Developed cutting edge neural networks with TensorFlow for Parkinson's detection using tremor data.

RESEARCH INTERESTS

Deep Learning, Efficient Neural Network Architectures, Transformer Architectures, Analog Neural Networks, Computer Vision, Resource-constrained ML Systems, Edge AI Solutions, Parallel and Distributed Computing, MLOps and Model Deployment

EDUCATION

- **Rutgers University, New Brunswick, NJ, (2019–2025 [Expected])**
  PhD Candidate & Master of Science (MS), Electrical and Computer Engineering (ECE)
  Grade Point Average (GPA): 3.92/4; ***Advised by*** **Dr. Dario Pompili**
- **Lahore University of Management Sciences (LUMS), Lahore, Pakistan (2015–2019)**
  Bachelor of Science (BS), Electrical Engineering (EE) – Gold Medalist
  Advised by Dr. Muhammad Tahir and Dr. Momin Uppal; Grade Point Average (GPA): 3.86/4

SELECTED RESEARCH PROJECTS

- **Synthetic Knee X-ray Generation — 2024**
  Developed a diffusion-based generative model to synthesize realistic knee X-ray images with controllable osteoarthritis severity levels. Achieved high-fidelity image synthesis with clinically relevant anatomical accuracy.
- **Multi-Agent Drone 3D Mapping with Reinforcement Learning — 2022**
  Designed and implemented a multi-agent reinforcement learning system for coordinated 3D object mapping using autonomous UAVs.
- **Ultra-Low Power Analog Neural Network Design for Health Monitoring —2023/24**
  Developed novel analog neural networks for ECG/EEG processing achieving micro-watt power consumption, published in IEEE JSAC and JBHI. Led NSF I-Corps customer discovery with 20 industry interviews to validate clinical needs.
- **Deep Joint Source Channel Coding for Underwater Image & Video Transmission — 2022**
  Developed a model using PyTorch for integrating deep learning models into complex underwater communication systems, demonstrated efficacy and published in world-class ACM conferences.
- **Crowd Prediction and Behavior Assessment Using Adaptive UAVs — 2024**
  Leveraged efficient signal processing techniques and polygonal flow estimation algorithms to predict real-time high-density crowd patterns from UAVs using multimodal sensor data

PATENTS

- **WO 2025/064997:** Anisotropic Diffusion-based Analog Neural Network Architecture
- **WO 2024/107672:** Techniques for Image Transmission through Acoustic Channels in Underwater Environments
- **(Pending Application):** Methods and Systems for Determining Group Motion Patterns

RELEVANT SKILLS

- **Programming Languages**: Python (advanced), C++, CUDA, MATLAB, Bash, SPICE
- **ML & AI Tools**: PyTorch, TensorFlow, OpenCV, Pandas, Jupyter, MLOps, Tensorboard, Llama 2, Scikit-learn
- **Edge Computing & DSP**: GNU Radio, ROS, low-power analog-digital ML systems
- **Cloud & DevOps**: Google Cloud Platform (GCP), Docker, GitHub Actions, Kubernetes
- **Deployment Environments**: Linux, Windows, FPGA, Embedded Systems (real-time AI)
- **Development Tools**: Visual Studio Code, Git, GitHub, Cursor

ACHIEVEMENTS

- Recipient of **Paul Panayotatos Scholarship** for Sustainability-aware Electrical Engineering, 2024
- Selected for **NSF NOVUS & NSF I-Corps** programs for innovation & customer discovery training, 2022
- Awarded **Travel Grants** from Rutgers University and ACM for **ACM WUWNet** and **IEEE MASS** participation
- Recognized with **Gold Medal** at LUMS for academic excellence, 2019
- Win: **TA Achievement Award**, 2020 from Rutgers